



in collaboration with



Linux Fundamental

ST5003CMD Operating Systems, Security, and Networks

*BSc (Hons) Computer Science with AI Softwarica College of IT &
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1. File Operation

1.1 Creating and navigating to the lab directory

I created a directory called 'linux_lab' and moved into it. This gave me a separate location to keep the file used during the lab.

```
root@AYUSHA:~# mkdir ~/linux_lab  
root@AYUSHA:~# cd ~/linux_lab
```

Figure 1 : Creating and navigating to the linux_lab directory

1.2 Create and Edit a Text File

I created a text file named 'greeting.txt' using Nano. After entering the required text, I saved the file and exited the editor. I then used the cat command to display and verify the contents of the file. I then used 'cat' to check the contents of the file.

```
root@AYUSHA:~/linux_lab# nano greeting.txt  
root@AYUSHA:~/linux_lab# cat greeting.txt  
Hello Linux Lab!
```

Figure 2 : Creating the greeting.txt file

1.3 Copy and Rename a File

The 'cp' command was used to create a copy of the file, and 'mv' was used to rename it. These commands are useful for basic file management from the terminal.

```
root@AYUSHA:~/linux_lab# cp greeting.txt backup.txt  
root@AYUSHA:~/linux_lab# mv backup.txt greeting_backup.txt  
root@AYUSHA:~/linux_lab# ls  
greeting.txt  greeting_backup.txt
```

Figure 3 : Creating and navigating to the linux_lab directory

1.4 Remove Files and Directories

I used 'rm' to remove a file and 'rmdir' to remove an empty directory. The 'rm -r' option can be used when a directory contains files.

```
root@AYUSHA:~/linux_lab# mkdir temp_dir
root@AYUSHA:~/linux_lab# ^C
root@AYUSHA:~/linux_lab# touch temp_dir/test.txt
root@AYUSHA:~/linux_lab# ls -l temp_dir
total 0
-rw-r--r-- 1 root root 0 Sep  8 06:09 test.txt
root@AYUSHA:~/linux_lab# rm -r temp_dir
root@AYUSHA:~/linux_lab# ls
greeting.txt  greeting_backup.txt
```

Figure 4 : Renaming file and removing an empty directory

1.5 Check File Permissions

Finally I checked the files with 'ls -l greeting.txt'. This displays information such as file permissions, ownership, size, and modification time.

```
root@AYUSHA:~/linux_lab# ls -l greeting.txt
-rw-r--r-- 1 root root 17 Sep  8 06:06 greeting.txt
root@AYUSHA:~/linux_lab#
```

Figure 5 : Checking File Permission

2. APT Package Management

2.1 Update the package List

Before installing the required development tools, I updated the package information using APT.

`sudo apt update`

```
root@AYUSHA:~/linux_lab# sudo apt update
Hit:1 http://archive.ubuntu.com/ubuntu resolute InRelease
Get:2 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 k
Get:3 http://archive.ubuntu.com/ubuntu resolute-updates InRelease [137 k
Get:4 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Com
nts [46.6 kB]
Get:5 http://archive.ubuntu.com/ubuntu resolute-backports InRelease [137
Get:6 http://security.ubuntu.com/ubuntu resolute-security/universe amd64
ponents [43.4 kB]
Get:7 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Packa
[601 kB]
Get:8 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Compo
s [98.3 kB]
Get:9 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 P
ges [268 kB]
Get:10 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64
onents [188 kB]
Get:11 http://archive.ubuntu.com/ubuntu resolute-updates/restricted amd6
ckages [409 kB]
Get:12 http://archive.ubuntu.com/ubuntu resolute-updates/restricted Tran
ion-en [79.5 kB]
Get:13 http://archive.ubuntu.com/ubuntu resolute-updates/multiverse amd6
ckages [11.6 kB]
Get:14 http://archive.ubuntu.com/ubuntu resolute-updates/multiverse Tran
ion-en [3036 B]
Get:15 http://archive.ubuntu.com/ubuntu resolute-backports/universe amd6
mponents [1056 B]
Fetched 2161 kB in 3s (819 kB/s)
42 packages can be upgraded. Run 'apt list --upgradable' to see them.
```

Figure 6 : Updating the package list

2.2 Install the C Compiler

I installed 'build-essential', which provides the main tools required for compiling C programs.

`sudo apt install build-essential`

```
root@AYUSHA:~/linux_lab# sudo apt install build-essential
Upgrading:
  libbz2-1.0

Installing:
  build-essential

Installing dependencies:
  bzip2                      libalgorithm-merge-perl
  dpkg-dev                  libcrypt-dev
  fakeroot                  libdpkg-perl
  g++                       libfakeroot
  g++-15                    libfile-fcntllock-perl
  g++-15-x86-64-linux-gnu  libstdc++-15-dev
  g++-x86-64-linux-gnu     lto-disabled-list
  libalgorithm-diff-perl   make
  libalgorithm-diff-xs-perl

Suggested packages:
  bzip2-doc                g++-multilib          bzip
  debian-keyring           g++-15-multilib      libstdc++-15-doc
  debian-tag2upload-keyring gcc-15-doc             make-doc

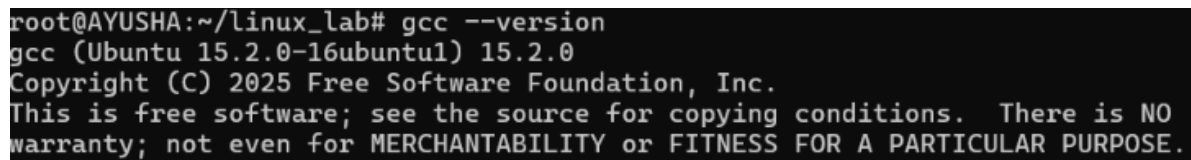
Summary:
  Upgrading: 1, Installing: 18, Removing: 0, Not Upgrading: 41
  Download size: 19.4 MB
  Space needed: 73.4 MB / 1024 GB available
```

Figure 7: Installing Build Essentials

2.3 Verify GCC Installation

I checked the installed GCC version to confirm that the compiler was available on the system.

`gcc --version`

A terminal window with a black background and white text. The prompt is 'root@AYUSHA:~/linux_lab#'. The command 'gcc --version' has been executed, and the output is displayed on the following lines: 'gcc (Ubuntu 15.2.0-16ubuntu1) 15.2.0', 'Copyright (C) 2025 Free Software Foundation, Inc.', 'This is free software; see the source for copying conditions. There is NO', and 'warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.'

```
root@AYUSHA:~/linux_lab# gcc --version
gcc (Ubuntu 15.2.0-16ubuntu1) 15.2.0
Copyright (C) 2025 Free Software Foundation, Inc.
This is free software; see the source for copying conditions. There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
```

Figure 8 : Checking the GCC Version

3. Compilation

3.1 Create the C Source File

I created a C source file called hello.c using Nano. The program prints a greeting and calculates the sum of two integer values.

```
root@AYUSHA:~/linux_lab# nano hello.c
root@AYUSHA:~/linux_lab# |
```

Figure 9 : Creating the C Source File

```
GNU nano 8.7.1 hello.c *
#include <stdio.h>

int main() {
    int a = 10, b = 5;

    printf("Hello, Linux Lab!\n");
    printf("Sum of %d and %d = %d\n", a, b, a + b);

    return 0;
}
```

Figure 10 : C Source Code for Basic Arithmetic Operations

3.2 Compile the Program

After saving the source code, I compiled it with GCC and named the resulting executable hello_program.

```
root@AYUSHA:~/linux_lab# gcc hello.c -o hello_program
root@AYUSHA:~/linux_lab# ls -l
total 28
-rw-r--r-- 1 root root  17 Sep  8 06:06 greeting.txt
-rw-r--r-- 1 root root  17 Sep  8 06:07 greeting_backup.txt
-rw-r--r-- 1 root root 161 Sep  8 06:23 hello.c
-rwxr-xr-x 1 root root 15992 Sep  8 06:24 hello_program
root@AYUSHA:~/linux_lab# |
```

Figure 11 : Compiling the C program

3.3 Check the Executable

I checked whether the executable had been created successfully using 'ls -l'.

ls -l hello_program

```
root@AYUSHA:~/linux_lab# ./hello_program
Hello, Linux Lab!
Sum of 10 and 5 = 15
root@AYUSHA:~/linux_lab# |
```

Figure 12 :Checking the compiled executable

3.5 Compile with Warnings

I also compiled the program using the '-Wall' option. This enables additional compiler warnings and can help identify potential problems in the code.

```
root@AYUSHA:~/linux_lab# gcc -Wall hello.c -o hello_program_warn
root@AYUSHA:~/linux_lab# ./hello_program_warn
Hello, Linux Lab!
Sum of 10 and 5 = 15
root@AYUSHA:~/linux_lab# |
```

Figure13 :Compiling with Warnings

4. Modify and Recompile

I changed the values in the C program and compiled it again. The program was then executed to check that the new values were reflected in the output.



```
GNU nano 8.7.1          hello.c *
#include <stdio.h>

int main() {
    int a = 100, b = 200;

    printf("Hello, Linux Lab!\n");
    printf("Sum of %d and %d = %d\n", a, b, a + b);

    return 0;
}
```

^G Help **^O Write Out** **^F Where Is** **^K Cut** **^T Execute**
^X Exit **^R Read File** **^_ Replace** **^U Paste** **^J Justify**

Figure14: Modifying the C Program

```
root@AYUSHA:~/linux_lab# nano hello.c
root@AYUSHA:~/linux_lab# cat hello.c
#include <stdio.h>

int main() {
    int a = 100, b = 200;

    printf("Hello, Linux Lab!\n");
    printf("Sum of %d and %d = %d\n", a, b, a + b);

    return 0;
}
root@AYUSHA:~/linux_lab# |
```

Figure15: Result of Modified C Program

```
root@AYUSHA:~/linux_lab# gcc -Wall hello.c -o hello_program
root@AYUSHA:~/linux_lab# ./hello_program
Hello, Linux Lab!
Sum of 100 and 200 = 300
root@AYUSHA:~/linux_lab# |
```

Figure16: Recompiling the Modifying Program and Verifying the updated output

5. Cleanup and Final Verification

After completing the lab, I returned to the home directory and checked the contents of the lab folder. I also displayed the C source file one final time. This final check confirmed that the source file and other lab files were stored in the expected directory.

```
root@AYUSHA:~/linux_lab# cd
root@AYUSHA:~# ls -lh ~/linux_lab
total 44K
-rw-r--r-- 1 root root  17 Sep  8 06:06 greeting.txt
-rw-r--r-- 1 root root  17 Sep  8 06:07 greeting_backup.txt
-rw-r--r-- 1 root root 164 Sep  8 06:32 hello.c
-rwxr-xr-x 1 root root 16K Sep  8 06:33 hello_program
-rwxr-xr-x 1 root root 16K Sep  8 06:27 hello_program_warn
root@AYUSHA:~# |
```

Figure17: Checking the Lab Directory

```
root@AYUSHA:~# cat ~/linux_lab/hello.c
#include <stdio.h>

int main() {
    int a = 100, b = 200;

    printf("Hello, Linux Lab!\n");
    printf("Sum of %d and %d = %d\n", a, b, a + b);

    return 0;
}
```

Figure18: Verifying the C Source Code